Proper base change for derived-complete pro-etale complexes

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Editorial note. Complete original author statement and proof retained with all source-local comparison/completion/pushforward support. Classical etale proper base change and formal adjunction map construction remain explicitly cited standard prerequisites. Original I-adic torsion and perfectness hypotheses retained. External chapter references display original labels without guessed numbers; shared pinned bibliography supplied. No prior local-model object counted anew.

Allowed standard prerequisites (editorial).

- Derived inverse limits, completed pullback, projection formula and classical etale proper base change. Standard formal construction of the base-change map and classical cited proper base change remain prerequisites. All source-local completion/Noetherian pushforward/comparison arguments are retained; no dropped perfectness assumption or collector proof.

1 Comparison with the étale site

Let X be a scheme. With suitable choices of sites[1] the functor $u : X_{étale} \to X_{pro\text{-}étale}$ sending U/X to U/X defines a morphism of sites

$$\epsilon : X_{pro\text{-}étale} \longrightarrow X_{étale}$$

This follows from Sites, Proposition [original label: `sites-proposition-get-morphism`].

Lemma 1.1. With notation as above. Let $\mathcal{F}$ be a sheaf on $X_{étale}$. The rule

$$X_{pro\text{-}étale} \longrightarrow \text{Sets}, \quad (f : Y \to X) \longmapsto \Gamma(Y_{étale}, f_{étale}^{-1}\mathcal{F})$$

is a sheaf and is equal to $\epsilon^{-1}\mathcal{F}$. Here $f_{étale} : Y_{étale} \to X_{étale}$ is the morphism of small étale sites constructed in Étale Cohomology, Section [original label: *`etale-cohomology-section-functoriality`*].

Proof. By Lemma [original label: `lemma-recognize-proetale-covering`] any pro-étale covering is an fpqc covering. Hence the formula defines a sheaf on $X_{pro\text{-}étale}$ by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-describe-pullback`]. Let $a : \mathrm{Sh}(X_{étale}) \to \mathrm{Sh}(X_{pro\text{-}étale})$ be the functor sending $\mathcal{F}$ to the sheaf given by the formula in the lemma. To show that $a = \epsilon^{-1}$ it suffices to show that a is a left adjoint to ϵ_*.

Let $\mathcal{G}$ be an object of $\mathrm{Sh}(X_{pro\text{-}étale})$. Recall that $\epsilon_*\mathcal{G}$ is simply given by the restriction of $\mathcal{G}$ to the full subcategory $X_{étale}$. Let $f : Y \to X$ be an object of $X_{pro\text{-}étale}$. We view $Y_{étale}$ as a subcategory of $X_{pro\text{-}étale}$. The restriction maps of the sheaf $\mathcal{G}$ define a map

$$\epsilon_*\mathcal{G} = \mathcal{G}|_{X_{étale}} \longrightarrow f_{étale,*}(\mathcal{G}|_{Y_{étale}})$$

Namely, for U in $X_{étale}$ the value of $f_{étale,*}(\mathcal{G}|_{Y_{étale}})$ on U is $\mathcal{G}(Y \times_X U)$ and there is a restriction map $\mathcal{G}(U) \to \mathcal{G}(Y \times_X U)$. By adjunction this determines a map

$$f_{étale}^{-1}(\epsilon_*\mathcal{G}) \to \mathcal{G}|_{Y_{étale}}$$

Putting these together for all $f : Y \to X$ in $X_{pro\text{-}étale}$ we obtain a canonical map $a(\epsilon_*\mathcal{G}) \to \mathcal{G}$.

Let $\mathcal{F}$ be an object of $\mathrm{Sh}(X_{étale})$. It is immediately clear that $\mathcal{F} = \epsilon_* a(\mathcal{F})$.

[1] Choose a big pro-étale site $\mathrm{Sch}_{pro\text{-}étale}$ containing X as in Definition [original label: `definition-big-proetale-site`]. Then let $\mathrm{Sch}_{étale}$ be the site having the same underlying category as $\mathrm{Sch}_{pro\text{-}étale}$ but whose coverings are exactly those pro-étale coverings which are also étale coverings. With these choices let $X_{étale}$ and $X_{pro\text{-}étale}$ be the subcategories defined in Definition [original label: `definition-big-small-proetale`] and Topologies, Definition [original label: `topologies-definition-big-small-etale`]. Compare with Topologies, Remark [original label: `topologies-remark-choice-sites`].

We claim the maps $\mathcal{F} \to \epsilon_* a(\mathcal{F})$ and $a(\epsilon_* \mathcal{G}) \to \mathcal{G}$ are the unit and counit of the adjunction (see Categories, Section [original label: categories-section-adjoint]). To see this it suffices to show that the corresponding maps

$$\mathrm{Mor}_{\mathrm{Sh}(X_{pro\text{-}étale})}(a(\mathcal{F}), \mathcal{G}) \to \mathrm{Mor}_{\mathrm{Sh}(X_{étale})}(\mathcal{F}, \epsilon^{-1}\mathcal{G})$$

and

$$\mathrm{Mor}_{\mathrm{Sh}(X_{étale})}(\mathcal{F}, \epsilon^{-1}\mathcal{G}) \to \mathrm{Mor}_{\mathrm{Sh}(X_{pro\text{-}étale})}(a(\mathcal{F}), \mathcal{G})$$

are mutually inverse. We omit the detailed verification. □

Lemma 1.2. Let X be a scheme. For every sheaf $\mathcal{F}$ on $X_{étale}$ the adjunction map $\mathcal{F} \to \epsilon_* \epsilon^{-1} \mathcal{F}$ is an isomorphism, i.e., $\epsilon^{-1}\mathcal{F}(U) = \mathcal{F}(U)$ for U in $X_{étale}$.

Proof. Follows immediately from the description of ϵ^{-1} in Lemma 1.1. □

Lemma 1.3. Let X be a scheme. Let $Y = \lim Y_i$ be the limit of a directed inverse system of quasi-compact and quasi-separated objects of $X_{pro\text{-}étale}$ with affine transition morphisms. For any sheaf $\mathcal{F}$ on $X_{étale}$ we have

$$\epsilon^{-1}\mathcal{F}(Y) = \operatorname{colim} \epsilon^{-1}\mathcal{F}(Y_i)$$

Moreover, if Y_i is in $X_{étale}$ we have $\epsilon^{-1}\mathcal{F}(Y) = \operatorname{colim} \mathcal{F}(Y_i)$.

Proof. By the description of $\epsilon^{-1}\mathcal{F}$ in Lemma 1.1, the displayed formula is a special case of Étale Cohomology, Theorem [original label: etale-cohomology-theorem-colimit]. (When X, Y, and the Y_i are all affine, see the easier to parse Étale Cohomology, Lemma [original label: etale-cohomology-lemma-directed-colimit-cohomology].) The final statement follows immediately from this and Lemma 1.2. □

Lemma 1.4. Let X be an affine scheme. For injective abelian sheaf $\mathcal{I}$ on $X_{étale}$ we have $H^p(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{I}) = 0$ for $p > 0$.

Proof. We are going to use Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-cech-vanish-collection] to prove this. Let $\mathcal{B} \subset \mathrm{Ob}(X_{pro\text{-}étale})$ be the set of affine objects U of $X_{pro\text{-}étale}$ such that $\mathcal{O}(X) \to \mathcal{O}(U)$ is ind-étale. Let Cov be the set of pro-étale coverings $\{U_i \to U\}_{i=1,\ldots,n}$ with $U \in \mathcal{B}$ such that $\mathcal{O}(U) \to \mathcal{O}(U_i)$ is ind-étale for $i = 1, \ldots, n$. Properties (1) and (2) of Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-cech-vanish-collection] hold for $\mathcal{B}$ and Cov by Lemmas [original label: lemma-composition-ind-etale], [original label: lemma-base-change-ind-etale], and [original label: lemma-proetale-affine] and Proposition [original label: proposition-weakly-etale].

To check condition (3) suppose that $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ is an element of Cov. We have to show that the higher Cech cohomology groups of $\epsilon^{-1}\mathcal{I}$ with respect to $\mathcal{U}$ are zero. First we write $U_i = \lim_{a \in A_i} U_{i,a}$ as a directed inverse limit with $U_{i,a} \to U$ étale and $U_{i,a}$ affine. We think of $A_1 \times \ldots \times A_n$ as a direct set with ordering $(a_1, \ldots, a_n) \geq (a'_1, \ldots, a'_n)$ if and only if $a_i \geq a'_i$ for $i = 1, \ldots, n$. Observe that $\mathcal{U}_{(a_1,\ldots,a_n)} = \{U_{i,a_i} \to U\}_{i=1,\ldots,n}$ is an étale covering for all $a_1, \ldots, a_n \in A_1 \times \ldots \times A_n$. Observe that

$$U_{i_0} \times_U U_{i_1} \times_U \ldots \times_U U_{i_p} = \lim_{(a_1,\ldots,a_n) \in A_1 \times \ldots \times A_n} U_{i_0,a_{i_0}} \times_U U_{i_1,a_{i_1}} \times_U \ldots \times_U U_{i_p,a_{i_p}}$$

for all $i_0, \ldots, i_p \in \{1, \ldots, n\}$ because limits commute with fibred products. Hence by Lemma 1.3 and exactness of filtered colimits we have

$$\check{H}^p(\mathcal{U}, \epsilon^{-1}\mathcal{I}) = \operatorname{colim} \check{H}^p(\mathcal{U}_{(a_1,\ldots,a_n)}, \epsilon^{-1}\mathcal{I})$$

Thus it suffices to prove the vanishing for étale coverings of U!

Let $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ be an étale covering with U_i affine. Write $U = \lim_{b \in B} U_b$ as a directed inverse limit with U_b affine and $U_b \to X$ étale. By Limits, Lemmas [original label: limits-lemma-descend-finite-presentation], [original label: limits-lemma-limit-affine], and [original label: limits-lemma-descend-etale] we can choose a $b_0 \in B$ such that for $i = 1, \ldots, n$ there is an étale morphism $U_{i,b_0} \to U_{b_0}$ of affines such that $U_i = U \times_{U_{b_0}} U_{i,b_0}$. Set $U_{i,b} = U_b \times_{U_{b_0}} U_{i,b_0}$ for $b \geq b_0$. For b large enough the family $\mathcal{U}_b = \{U_{i,b} \to U_b\}_{i=1,\ldots,n}$ is an étale covering, see Limits, Lemma [original label: limits-lemma-descend-surjective]. Exactly as before we find that

$$\check{H}^p(\mathcal{U}, \epsilon^{-1}\mathcal{I}) = \operatorname{colim} \check{H}^p(\mathcal{U}_b, \epsilon^{-1}\mathcal{I}) = \operatorname{colim} \check{H}^p(\mathcal{U}_b, \mathcal{I})$$

the final equality by Lemma 1.2. Since each of the Čech complexes on the right hand side is acyclic in positive degrees (Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-injective-trivial-cech]) it follows that the one on the left is too. This proves condition (3) of Cohomology on Sites, Lemma [original label: sites-cohomology-lemma-cech-vanish-collection]. Since $X \in \mathcal{B}$ the lemma follows. □

Lemma 1.5. Let X be a scheme.

1. For an abelian sheaf $\mathcal{F}$ on $X_{étale}$ we have $R\epsilon_*(\epsilon^{-1}\mathcal{F}) = \mathcal{F}$.
2. For $K \in D^+(X_{étale})$ the map $K \to R\epsilon_*\epsilon^{-1}K$ is an isomorphism.

Proof. Let $\mathcal{I}$ be an injective abelian sheaf on $X_{étale}$. Recall that $R^q\epsilon_*(\epsilon^{-1}\mathcal{I})$ is the sheaf associated to $U \mapsto H^q(U_{pro\text{-}étale}, \epsilon^{-1}\mathcal{I})$, see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-higher-direct-images`]. By Lemma 1.4 we see that this is zero for $q > 0$ and U affine and étale over X. Since every object of $X_{étale}$ has a covering by affine objects, it follows that $R^q\epsilon_*(\epsilon^{-1}\mathcal{I}) = 0$ for $q > 0$.

Let $K \in D^+(X_{étale})$. Choose a bounded below complex $\mathcal{I}^\bullet$ of injective abelian sheaves on $X_{étale}$ representing K. Then $\epsilon^{-1}K$ is represented by $\epsilon^{-1}\mathcal{I}^\bullet$. By Leray's acyclicity lemma (Derived Categories, Lemma [original label: `derived-lemma-leray-acyclicity`]) we see that $R\epsilon_*\epsilon^{-1}K$ is represented by $\epsilon_*\epsilon^{-1}\mathcal{I}^\bullet$. By Lemma 1.2 we conclude that $R\epsilon_*\epsilon^{-1}\mathcal{I}^\bullet = \mathcal{I}^\bullet$ and the proof of (2) is complete. Part (1) is a special case of (2). □

Lemma 1.6. Let X be a scheme.

1. For an abelian sheaf $\mathcal{F}$ on $X_{étale}$ we have
$$H^i(X_{étale}, \mathcal{F}) = H^i(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{F})$$
for all i.
2. For $K \in D^+(X_{étale})$ we have
$$R\Gamma(X_{étale}, K) = R\Gamma(X_{pro\text{-}étale}, \epsilon^{-1}K)$$

Proof. Immediate consequence of Lemma 1.5 and the Leray spectral sequence (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-apply-Leray`]). □

Lemma 1.7. Let X be a scheme. Let $\mathcal{G}$ be a sheaf of (possibly noncommutative) groups on $X_{étale}$. We have
$$H^1(X_{étale}, \mathcal{G}) = H^1(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{G})$$
where H^1 is defined as the set of isomorphism classes of torsors (see Cohomology on Sites, Section [original label: *`sites-cohomology-section-h1-torsors`*]).

Proof. Since the functor ϵ^{-1} is fully faithful by Lemma 1.2 it is clear that the map $H^1(X_{étale}, \mathcal{G}) \to H^1(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{G})$ is injective. To show surjectivity it suffices to show that any $\epsilon^{-1}\mathcal{G}$-torsor $\mathcal{F}$ is étale locally trivial. To do this we may assume that X is affine. Thus we reduce to proving surjectivity for X affine.

Choose a covering $\{U \to X\}$ with (a) U affine, (b) $\mathcal{O}(X) \to \mathcal{O}(U)$ ind-étale, and (c) $\mathcal{F}(U)$ nonempty. We can do this by Proposition [original label: `proposition-weakly-etale`] and the fact that standard pro-étale coverings of X are cofinal among all pro-étale coverings of X (Lemma [original label: `lemma-proetale-affine`]). Write $U = \lim U_i$ as a limit of affine schemes étale over X. Pick $s \in \mathcal{F}(U)$. Let $g \in \epsilon^{-1}\mathcal{G}(U \times_X U)$ be the unique section such that $g \cdot \mathrm{pr}_1^* s = \mathrm{pr}_2^* s$ in $\mathcal{F}(U \times_X U)$. Then g satisfies the cocycle condition
$$\mathrm{pr}_{12}^* g \cdot \mathrm{pr}_{23}^* g = \mathrm{pr}_{13}^* g$$
in $\epsilon^{-1}\mathcal{G}(U \times_X U \times_X U)$. By Lemma 1.3 we have
$$\epsilon^{-1}\mathcal{G}(U \times_X U) = \mathrm{colim}\, \mathcal{G}(U_i \times_X U_i)$$
and
$$\epsilon^{-1}\mathcal{G}(U \times_X U \times_X U) = \mathrm{colim}\, \mathcal{G}(U_i \times_X U_i \times_X U_i)$$
hence we can find an i and an element $g_i \in \mathcal{G}(U_i \times_X U_i)$ mapping to g satisfying the cocycle condition. The cocycle g_i then defines a torsor for $\mathcal{G}$ on $X_{étale}$ whose pullback is isomorphic to $\mathcal{F}$ by construction. Some details omitted (namely, the relationship between torsors and 1-cocycles which should be added to the chapter on cohomology on sites). □

Lemma 1.8. Let X be a scheme. Let Λ be a ring.

1. The essential image of the fully faithful functor $\epsilon^{-1} : \mathrm{Mod}(X_{étale}, \Lambda) \to \mathrm{Mod}(X_{pro\text{-}étale}, \Lambda)$ is a weak Serre subcategory $\mathcal{C}$.
2. The functor ϵ^{-1} defines an equivalence of categories of $D^+(X_{étale}, \Lambda)$ with $D^+_{\mathcal{C}}(X_{pro\text{-}étale}, \Lambda)$ with quasi-inverse given by $R\epsilon_*$.

Proof. To prove (1) we will prove conditions (1) – (4) of Homology, Lemma [original label: `homology-lemma-characterize-weak-serre-subcategory`]. Since ϵ^{-1} is fully faithful (Lemma 1.2) and exact, everything is clear except for condition (4). However, if

$$0 \to \epsilon^{-1}\mathcal{F}_1 \to \mathcal{G} \to \epsilon^{-1}\mathcal{F}_2 \to 0$$

is a short exact sequence of sheaves of Λ-modules on $X_{pro\text{-}étale}$, then we get

$$0 \to \epsilon_*\epsilon^{-1}\mathcal{F}_1 \to \epsilon_*\mathcal{G} \to \epsilon_*\epsilon^{-1}\mathcal{F}_2 \to R^1\epsilon_*\epsilon^{-1}\mathcal{F}_1$$

which by Lemma 1.5 is the same as a short exact sequence

$$0 \to \mathcal{F}_1 \to \epsilon_*\mathcal{G} \to \mathcal{F}_2 \to 0$$

Pulling pack we find that $\mathcal{G} = \epsilon^{-1}\epsilon_*\mathcal{G}$. This proves (1).

Part (2) follows from part (1) and Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-equivalence-bounded`]. □

Let Λ be a ring. In Modules on Sites, Section [original label: `sites-modules-section-locally-constant`] we have defined the notion of a locally constant sheaf of Λ-modules on a site. If M is a Λ-module, then $\underline{M}$ is of finite presentation as a sheaf of $\underline{\Lambda}$-modules if and only if M is a finitely presented Λ-module, see Modules on Sites, Lemma [original label: `sites-modules-lemma-locally-constant-finite-type`].

Lemma 1.9. Let X be a scheme. Let Λ be a ring. The functor ϵ^{-1} defines an equivalence of categories

$$\left\{\begin{array}{c}\text{locally constant sheaves}\\ \text{of } \Lambda\text{-modules on } X_{étale}\\ \text{of finite presentation}\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{locally constant sheaves}\\ \text{of } \Lambda\text{-modules on } X_{pro\text{-}étale}\\ \text{of finite presentation}\end{array}\right\}$$

Proof. Let $\mathcal{F}$ be a locally constant sheaf of Λ-modules on $X_{pro\text{-}étale}$ of finite presentation. Choose a pro-étale covering $\{U_i \to X\}$ such that $\mathcal{F}|_{U_i}$ is constant, say $\mathcal{F}|_{U_i} \cong \underline{M_i}_{U_i}$. Observe that $U_i \times_X U_j$ is empty if M_i is not isomorphic to M_j. For each Λ-module M let $I_M = \{i \in I \mid M_i \cong M\}$. As pro-étale coverings are fpqc coverings and by Descent, Lemma [original label: `descent-lemma-open-fpqc-covering`] we see that $U_M = \bigcup_{i \in I_M} \mathrm{Im}(U_i \to X)$ is an open subset of X. Then $X = \coprod U_M$ is a disjoint open covering of X. We may replace X by U_M for some M and assume that $M_i = M$ for all i.

Consider the sheaf $\mathcal{I} = \mathrm{Isom}(\underline{M}, \mathcal{F})$. This sheaf is a torsor for $\mathcal{G} = \mathrm{Isom}(\underline{M}, \underline{M})$. By Modules on Sites, Lemma [original label: `sites-modules-lemma-locally-constant`] we have $\mathcal{G} = \underline{G}$ where $G = \mathrm{Isom}_\Lambda(M, M)$. Since torsors for the étale topology and the pro-étale topology agree by Lemma 1.7 it follows that $\mathcal{I}$ has sections étale locally on X. Thus $\mathcal{F}$ is étale locally a constant sheaf which is what we had to show. □

Lemma 1.10. Let X be a scheme. Let Λ be a Noetherian ring. Let $D_{flc}(X_{étale}, \Lambda)$, resp. $D_{flc}(X_{pro\text{-}étale}, \Lambda)$ be the full subcategory of $D(X_{étale}, \Lambda)$, resp. $D(X_{pro\text{-}étale}, \Lambda)$ consisting of those complexes whose cohomology sheaves are locally constant sheaves of Λ-modules of finite type. Then

$$\epsilon^{-1} : D^+_{flc}(X_{étale}, \Lambda) \longrightarrow D^+_{flc}(X_{pro\text{-}étale}, \Lambda)$$

is an equivalence of categories.

Proof. The categories $D_{flc}(X_{étale}, \Lambda)$ and $D_{flc}(X_{pro\text{-}étale}, \Lambda)$ are strictly full, saturated, triangulated subcategories of $D(X_{étale}, \Lambda)$ and $D(X_{pro\text{-}étale}, \Lambda)$ by Modules on Sites, Lemma [original label: `sites-modules-lemma-kernel-finite-locally-constant`] and Derived Categories, Section [original label: `derived-section-triangulated-sub`]. The statement of the lemma follows by combining Lemmas 1.8 and 1.9. □

Lemma 1.11. Let X be a scheme. Let Λ be a Noetherian ring. Let K be an object of $D(X_{pro\text{-}étale}, \Lambda)$. Set $K_n = K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$. If K_1 is

1. in the essential image of $\epsilon^{-1} : D(X_{étale}, \Lambda/I) \to D(X_{pro\text{-}étale}, \Lambda/I)$, and
2. has tor amplitude in $[a, \infty)$ for some $a \in \mathrm{Z}$,

then (1) and (2) hold for K_n as an object of $D(X_{pro\text{-}étale}, \Lambda/I^n)$.

Proof. Assertion (2) for K_n follows from the more general Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-bounded`]. Assertion (1) for K_n follows by induction on n from the distinguished triangles

$$K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}} \to K_{n+1} \to K_n \to K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}}[1]$$

and the isomorphism

$$K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}} = K_1 \otimes^{\mathrm{L}}_{\Lambda/I} \underline{I^n/I^{n+1}}$$

and the fact proven in Lemma 1.8 that the essential image of ϵ^{-1} is a triangulated subcategory of $D^+(X_{pro\text{-}étale}, \Lambda/I^n)$. □

Example 1.12. Let X be a scheme. Let A be an abelian group. Denote $fun(-, A)$ the sheaf on $X_{pro\text{-}étale}$ which maps U to the set of all maps $U \to A$ (of sets of points). Consider the sequence of sheaves

$$0 \to \underline{A} \to fun(-, A) \to \mathcal{F} \to 0$$

on $X_{pro\text{-}étale}$. Since the constant sheaf is the pullback from the final topos we see that $\underline{A} = \epsilon^{-1}\underline{A}$. However, if A has more than one element, then neither $fun(-, A)$ nor $\mathcal{F}$ are pulled back from the étale site of X. To work out the values of $\mathcal{F}$ in some cases, assume that all points of X are closed with separably closed residue fields and U is affine. Then all points of U are closed with separably closed residue fields and we have

$$H^1_{pro\text{-}étale}(U, \underline{A}) = H^1_{étale}(U, \underline{A}) = 0$$

by Lemma 1.6 and Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-affine-only-closed-points`]. Hence in this case we have

$$\mathcal{F}(U) = fun(U, A)/\underline{A}(U)$$

2 Derived completion in the constant Noetherian case

We continue the discussion started in Algebraic and Formal Geometry, Section [original label: `algebraization-section-derived-completion`]; we assume the reader has read at least some of that section.

Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Recall from Modules on Sites, Lemma [original label: `sites-modules-lemma-completion-flat`] that

$$\underline{\Lambda}^\wedge = \lim \underline{\Lambda/I^n}$$

is a flat $\underline{\Lambda}$-algebra and that the map $\underline{\Lambda} \to \underline{\Lambda}^\wedge$ identifies quotients by I. Hence Algebraic and Formal Geometry, Lemma [original label: `algebraization-lemma-restriction-derived-complete-equivalence`] tells us that

$$D_{comp}(\mathcal{C}, \Lambda) = D_{comp}(\mathcal{C}, \underline{\Lambda}^\wedge)$$

In particular the cohomology sheaves $H^i(K)$ of an object K of $D_{comp}(\mathcal{C}, \Lambda)$ are sheaves of $\underline{\Lambda}^\wedge$-modules. For notational convenience we often work with $D_{comp}(\mathcal{C}, \Lambda)$.

Lemma 2.1. Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. The left adjoint to the inclusion functor $D_{comp}(\mathcal{C}, \Lambda) \to D(\mathcal{C}, \Lambda)$ of Algebraic and Formal Geometry, Proposition [original label: *`algebraization-proposition-derived-completion`*] sends K to

$$K^\wedge = R\lim(K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n})$$

In particular, K is derived complete if and only if $K = R\lim(K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n})$.

Proof. Choose generators $f_1, \ldots, f_r$ of I. By Algebraic and Formal Geometry, Lemma [original label: `algebraization-lemma-derived-completion-koszul`] we have

$$K^\wedge = R\lim(K \otimes^{\mathrm{L}}_{\Lambda} \underline{K_n})$$

where $K_n = K(\Lambda, f_1^n, \ldots, f_r^n)$. In More on Algebra, Lemma [original label: `more-algebra-lemma-sequence-Koszul-complexes`] we have seen that the pro-systems $\{K_n\}$ and $\{\Lambda/I^n\}$ of $D(\Lambda)$ are isomorphic. Thus the lemma follows. □

Lemma 2.2. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let $f : \mathrm{Sh}(\mathcal{D}) \to \mathrm{Sh}(\mathcal{C})$ be a morphism of topoi. Then

1. Rf_* sends $D_{comp}(\mathcal{D}, \Lambda)$ into $D_{comp}(\mathcal{C}, \Lambda)$,
2. the map $Rf_* : D_{comp}(\mathcal{D}, \Lambda) \to D_{comp}(\mathcal{C}, \Lambda)$ has a left adjoint $Lf^*_{comp} : D_{comp}(\mathcal{C}, \Lambda) \to D_{comp}(\mathcal{D}, \Lambda)$ which is Lf^* followed by derived completion,
3. Rf_* commutes with derived completion,
4. for K in $D_{comp}(\mathcal{D}, \Lambda)$ we have $Rf_*K = R\lim Rf_*(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$.
5. for M in $D_{comp}(\mathcal{C}, \Lambda)$ we have $Lf^*_{comp}M = R\lim Lf^*(M \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$.

Proof. We have seen (1) and (2) in Algebraic and Formal Geometry, Lemma [original label: `algebraization-lemma-pushforward-derived-complete-adjoint`]. Part (3) follows from Algebraic and Formal Geometry, Lemma [original label: `algebraization-lemma-pushforward-commutes-with-derived-completion`]. For (4) let K be derived complete. Then

$$Rf_*K = Rf_*(R\lim K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}) = R\lim Rf_*(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$$

the first equality by Lemma 2.1 and the second because Rf_* commutes with $R\lim$ (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-Rf-commutes-with-Rlim`]). This proves (4). To prove (5), by Lemma 2.1 we have

$$Lf^*_{comp}M = R\lim(Lf^*M \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$$

Since Lf^* commutes with derived tensor product by Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-pullback-tensor-product`] and since $Lf^*\underline{\Lambda/I^n} = \underline{\Lambda/I^n}$ we get (5). □

3 Derived completion and weakly contractible objects

We continue the discussion in Section 2. In this section we will see how the existence of weakly contractible objects simplifies the study of derived complete modules.

Let $\mathcal{C}$ be a site. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Although the general theory concerning $D_{comp}(\mathcal{C}, \Lambda)$ is quite satisfactory it is hard to explicitly give examples of derived complete complexes. We know that

1. every object M of $D(\mathcal{C}, \Lambda/I^n)$ restricts to a derived complete object of $D(\mathcal{C}, \Lambda)$, and
2. for every $K \in D(\mathcal{C}, \Lambda)$ the derived completion $K^\wedge = R\lim(K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n})$ is derived complete.

The first type of objects are trivially complete and perhaps not interesting. The problem with (2) is that derived completion in general is somewhat mysterious, even in case $K = \underline{\Lambda}$. Namely, by definition of homotopy limits there is a distinguished triangle

$$R\lim(\underline{\Lambda/I^n}) \to \prod \underline{\Lambda/I^n} \to \prod \underline{\Lambda/I^n} \to R\lim(\underline{\Lambda/I^n})[1]$$

in $D(\mathcal{C}, \Lambda)$ where the products are in $D(\mathcal{C}, \Lambda)$. These are computed by taking products of injective resolutions (Injectives, Lemma [original label: `injectives-lemma-derived-products`]), so we see that the sheaf $H^p(\prod \underline{\Lambda/I^n})$ is the sheafification of the presheaf

$$U \longmapsto \prod H^p(U, \Lambda/I^n).$$

As an explicit example, if $X = \mathrm{Spec}(\mathrm{C}[t, t^{-1}])$, $\mathcal{C} = X_{étale}$, $\Lambda = \mathrm{Z}$, $I = (2)$, and $p = 1$, then we get the sheafification of the presheaf

$$U \mapsto \prod H^1(U_{étale}, \mathrm{Z}/2^n\mathrm{Z})$$

for U étale over X. Note that $H^1(X_{étale}, \mathrm{Z}/m\mathrm{Z})$ is cyclic of order m with generator α_m given by the finite étale $\mathrm{Z}/m\mathrm{Z}$-covering given by the equation $t = s^m$ (see Étale Cohomology, Section [original label: `etale-cohomology-section-computation`]). Then the section

$$\alpha = (\alpha_{2^n}) \in \prod H^1(X_{étale}, \mathrm{Z}/2^n\mathrm{Z})$$

of the presheaf above does not restrict to zero on any nonempty étale scheme over X, whence the sheaf associated to the presheaf is not zero.

However, on the pro-étale site this phenomenon does not occur. The reason is that we have enough (quasi-compact) weakly contractible objects. In the following proposition we collect some results about derived completion in the Noetherian constant case for sites having enough weakly contractible objects (see Sites, Definition [original label: `sites-definition-w-contractible`]).

Proposition 3.1. Let $\mathcal{C}$ be a site. Assume $\mathcal{C}$ has enough weakly contractible objects. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal.

1. The category of derived complete sheaves Λ-modules is a weak Serre subcategory of $\mathrm{Mod}(\mathcal{C}, \Lambda)$.
2. A sheaf $\mathcal{F}$ of Λ-modules satisfies $\mathcal{F} = \lim \mathcal{F}/I^n\mathcal{F}$ if and only if $\mathcal{F}$ is derived complete and $\bigcap I^n\mathcal{F} = 0$.
3. The sheaf $\underline{\Lambda}^\wedge$ is derived complete.
4. If $\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$ is an inverse system of derived complete sheaves of Λ-modules, then $\lim \mathcal{F}_n$ is derived complete.
5. An object $K \in D(\mathcal{C}, \Lambda)$ is derived complete if and only if each cohomology sheaf $H^p(K)$ is derived complete.
6. An object $K \in D_{comp}(\mathcal{C}, \Lambda)$ is bounded above if and only if $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ is bounded above.
7. An object $K \in D_{comp}(\mathcal{C}, \Lambda)$ is bounded if $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has finite tor dimension.

Proof. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset such that every $U \in \mathcal{B}$ is weakly contractible and every object of $\mathcal{C}$ has a covering by elements of $\mathcal{B}$. We will use the results of Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-w-contractible`] and Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`] without further mention.

Recall that $R\lim$ commutes with $R\Gamma(U, -)$, see Injectives, Lemma [original label: `injectives-lemma-RF-commutes-with-Rlim`]. Let $f \in I$. Recall that $T(K, f)$ is the homotopy limit of the system

$$\ldots \xrightarrow{f} K \xrightarrow{f} K \xrightarrow{f} K$$

in $D(\mathcal{C}, \Lambda)$. Thus

$$R\Gamma(U, T(K, f)) = T(R\Gamma(U, K), f).$$

Since we can test isomorphisms of maps between objects of $D(\mathcal{C}, \Lambda)$ by evaluating at $U \in \mathcal{B}$ we conclude an object K of $D(\mathcal{C}, \Lambda)$ is derived complete if and only if for every $U \in \mathcal{B}$ the object $R\Gamma(U, K)$ is derived complete as an object of $D(\Lambda)$.

The remark above implies that items (1), (5) follow from the corresponding results for modules over rings, see More on Algebra, Lemmas [original label: `more-algebra-lemma-hom-from-Af`] and [original label: `more-algebra-lemma-serre-subcategory`]. In the same way (2) can be deduced from More on Algebra, Proposition [original label: `more-algebra-proposition-derived-complete-modules`] as $(I^n\mathcal{F})(U) = I^n \cdot \mathcal{F}(U)$ for $U \in \mathcal{B}$ (by exactness of evaluating at U).

Proof of (4). The homotopy limit $R\lim \mathcal{F}_n$ is in $D_{comp}(X, \Lambda)$ (see discussion following Algebraic and Formal Geometry, Definition [original label: `algebraization-definition-derived-complete`]). By part (5) just proved we conclude that $\lim \mathcal{F}_n = H^0(R\lim \mathcal{F}_n)$ is derived complete. Part (3) is a special case of (4).

Proof of (6) and (7). Follows from Lemma 2.1 and Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-bounded`] and the computation of homotopy limits in Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`]. □

4 Cohomology of a point

Let Λ be a Noetherian ring complete with respect to an ideal $I \subset \Lambda$. Let k be a field. In this section we "compute"

$$H^i(\mathrm{Spec}(k)_{pro\text{-}étale}, \underline{\Lambda}^\wedge)$$

where $\underline{\Lambda}^\wedge = \lim_m \underline{\Lambda/I^m}$ as before. Let k^{sep} be a separable algebraic closure of k. Then

$$\mathcal{U} = \{\mathrm{Spec}(k^{sep}) \to \mathrm{Spec}(k)\}$$

is a pro-étale covering of $\mathrm{Spec}(k)$. We will use the Čech to cohomology spectral sequence with respect to this covering. Set $U_0 = \mathrm{Spec}(k^{sep})$ and

$$\begin{aligned} U_n &= \mathrm{Spec}(k^{sep}) \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k^{sep}) \times_{\mathrm{Spec}(k)} \ldots \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k^{sep}) \\ &= \mathrm{Spec}(k^{sep} \otimes_k k^{sep} \otimes_k \ldots \otimes_k k^{sep}) \end{aligned}$$

($n+1$ factors). Note that the underlying topological space $|U_0|$ of U_0 is a singleton and for $n \geq 1$ we have

$$|U_n| = G \times \ldots \times G \quad (n \text{ factors})$$

as profinite spaces where $G = \mathrm{Gal}(k^{sep}/k)$. Namely, every point of U_n has residue field k^{sep} and we identify $(\sigma_1, \ldots, \sigma_n)$ with the point corresponding to the surjection

$$k^{sep} \otimes_k k^{sep} \otimes_k \ldots \otimes_k k^{sep} \longrightarrow k^{sep}, \quad \lambda_0 \otimes \lambda_1 \otimes \ldots \lambda_n \longmapsto \lambda_0 \sigma_1(\lambda_1) \ldots \sigma_n(\lambda_n)$$

Then we compute

$$\begin{aligned} R\Gamma((U_n)_{pro\text{-}étale}, \underline{\Lambda}^\wedge) &= R\lim_m R\Gamma((U_n)_{pro\text{-}étale}, \underline{\Lambda/I^m}) \\ &= R\lim_m R\Gamma((U_n)_{étale}, \underline{\Lambda/I^m}) \\ &= \lim_m H^0(U_n, \underline{\Lambda/I^m}) \\ &= \mathrm{Maps}_{cont}(G \times \ldots \times G, \Lambda) \end{aligned}$$

The first equality because $R\Gamma$ commutes with derived limits and as $\Lambda^\wedge$ is the derived limit of the sheaves $\underline{\Lambda/I^m}$ by Proposition 3.1. The second equality by Lemma 1.6. The third equality by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-affine-only-closed-points`]. The fourth equality uses Étale Cohomology, Remark [original label: `etale-cohomology-remark-constant-locally-constant-maps`] to identify sections of the constant sheaf $\underline{\Lambda/I^m}$. Then it uses the fact that Λ is complete with respect to I and hence equal to $\lim_m \Lambda/I^m$ as a topological space, to see that $\lim_m \mathrm{Map}_{cont}(G, \Lambda/I^m) = \mathrm{Map}_{cont}(G, \Lambda)$ and similarly for higher powers of G. At this point Cohomology on Sites, Lemmas [original label: `sites-cohomology-lemma-cech-cohomology`] and [original label: `sites-cohomology-lemma-cech-spectral-sequence-application`] tell us that

$$\Lambda \to \mathrm{Maps}_{cont}(G, \Lambda) \to \mathrm{Maps}_{cont}(G \times G, \Lambda) \to \ldots$$

computes the pro-étale cohomology. In other words, we see that

$$H^i(\mathrm{Spec}(k)_{pro\text{-}étale}, \underline{\Lambda}^\wedge) = H^i_{cont}(G, \Lambda)$$

where the right hand side is Tate's continuous cohomology, see Étale Cohomology, Section [original label: `etale-cohomology-section-continuous-group-cohomology`]. Of course, this is as it should be.

Lemma 4.1. Let k be a field. Let $G = \mathrm{Gal}(k^{sep}/k)$ be its absolute Galois group. Further,

1. let M be a profinite abelian group with a continuous G-action, or
2. let Λ be a Noetherian ring and $I \subset \Lambda$ an ideal an let M be an I-adically complete Λ-module with continuous G-action.

Then there is a canonical sheaf $\underline{M}^\wedge$ on $\mathrm{Spec}(k)_{pro\text{-}étale}$ associated to M such that

$$H^i(\mathrm{Spec}(k), \underline{M}^\wedge) = H^i_{cont}(G, M)$$

as abelian groups or Λ-modules.

Proof. Proof in case (2). Set $M_n = M/I^n M$. Then $M = \lim M_n$ as M is assumed I-adically complete. Since the action of G is continuous we get continuous actions of G on M_n. By Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-equivalence-sheaves-point`] this action corresponds to a (locally constant) sheaf $\underline{M_n}$ of Λ/I^n-modules on $\mathrm{Spec}(k)_{étale}$. Pull back to $\mathrm{Spec}(k)_{pro\text{-}étale}$ by the comparison morphism ϵ and take the limit

$$\underline{M}^\wedge = \lim \epsilon^{-1} \underline{M_n}$$

to get the sheaf promised in the lemma. Exactly the same argument as given in the introduction of this section gives the comparison with Tate's continuous Galois cohomology. □

5 Functoriality of the pro-étale site

Let $f : X \to Y$ be a morphism of schemes. The functor $Y_{pro\text{-}étale} \to X_{pro\text{-}étale}$, $V \mapsto X \times_Y V$ induces a morphism of sites $f_{pro\text{-}étale} : X_{pro\text{-}étale} \to Y_{pro\text{-}étale}$, see Sites, Proposition [original label: `sites-proposition-get-morphism`]. In fact, we obtain a commutative diagram of morphisms of sites

$$\begin{array}{ccc} X_{pro\text{-}étale} & \xrightarrow{\epsilon} & X_{étale} \\ {\scriptstyle f_{pro\text{-}étale}}\downarrow & & \downarrow{\scriptstyle f_{étale}} \\ Y_{pro\text{-}étale} & \xrightarrow{\epsilon} & Y_{étale} \end{array}$$

where ϵ is as in Section 1. In particular we have $\epsilon^{-1} f^{-1}_{étale} = f^{-1}_{pro\text{-}étale} \epsilon^{-1}$. Here is the corresponding result for pushforward.

Lemma 5.1. Let $f : X \to Y$ be a morphism of schemes which is quasi-compact and quasi-separated.

1. Let $\mathcal{F}$ be a sheaf of sets on $X_{étale}$. Then we have $f_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F} = \epsilon^{-1}f_{étale,*}\mathcal{F}$.
2. Let $\mathcal{F}$ be an abelian sheaf on $X_{étale}$. Then we have $Rf_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F} = \epsilon^{-1}Rf_{étale,*}\mathcal{F}$.

Proof. Proof of (1). Let $\mathcal{F}$ be a sheaf of sets on $X_{étale}$. There is a canonical map $\epsilon^{-1}f_{étale,*}\mathcal{F} \to f_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F}$, see Sites, Section [original label: `sites-section-pullback`]. To show it is an isomorphism we may work (Zariski) locally on Y, hence we may assume Y is affine. In this case every object of $Y_{pro\text{-}étale}$ has a covering by objects $V = \lim V_i$ which are limits of affine schemes V_i étale over Y (by Proposition [original label: `proposition-weakly-etale`] for example). Evaluating the map $\epsilon^{-1}f_{étale,*}\mathcal{F} \to f_{pro\text{-}étale,*}\epsilon^{-1}\mathcal{F}$ on V we obtain a map

$$\operatorname{colim}\Gamma(X \times_Y V_i, \mathcal{F}) \longrightarrow \Gamma(X \times_Y V, \epsilon^*\mathcal{F})$$

See Lemma 1.3 for the left hand side. By Lemma 1.1 we have

$$\Gamma(X \times_Y V, \epsilon^*\mathcal{F}) = \Gamma(X \times_Y V, g_{étale}^{-1}\mathcal{F})$$

where $g : X \times_Y V \to X$ is the projection. Hence the result holds by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-directed-colimit-cohomology`].

Proof of (2). Arguing in exactly the same manner as above we see that it suffices to show that

$$\operatorname{colim} H^i_{étale}(X \times_Y V_i, \mathcal{F}) \longrightarrow H^i_{étale}(X \times_Y V, \mathcal{F})$$

which follows once more from Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-directed-colimit-cohomology`]. □

6 Finite morphisms and pro-étale sites

It is not clear that a finite morphism of schemes determines an exact pushforward on abelian pro-étale sheaves.

Lemma 6.1. Let $f : Z \to X$ be a finite morphism of schemes which is locally of finite presentation. Then $f_{pro\text{-}étale,*} : \mathrm{Ab}(Z_{pro\text{-}étale}) \to \mathrm{Ab}(X_{pro\text{-}étale})$ is exact.

Proof. The prove this we may work (Zariski) locally on X and assume that X is affine, say $X = \operatorname{Spec}(A)$. Then $Z = \operatorname{Spec}(B)$ for some finite A-algebra B of finite presentation. The construction in the proof of Proposition [original label: `proposition-find-w-contractible`] produces a faithfully flat, ind-étale ring map $A \to D$ with D w-contractible. We may check exactness of a sequence of sheaves by evaluating on $U = \operatorname{Spec}(D)$ be such an object. Then $f_{pro\text{-}étale,*}\mathcal{F}$ evaluated at U is equal to $\mathcal{F}$ evaluated at $V = \operatorname{Spec}(D \otimes_A B)$. Since $D \otimes_A B$ is w-contractible by Lemma [original label: `lemma-finite-finitely-presented-over-w-contractible`] evaluation at V is exact. □

7 Closed immersions and pro-étale sites

It is not clear (and likely false) that a closed immersion of schemes determines an exact pushforward on abelian pro-étale sheaves.

Lemma 7.1. Let $i : Z \to X$ be a closed immersion morphism of affine schemes. Denote X_{app} and Z_{app} the sites introduced in Lemma [original label: *`lemma-affine-alternative`*]. The base change functor

$$u : X_{app} \to Z_{app}, \quad U \longmapsto u(U) = U \times_X Z$$

is continuous and has a fully faithful left adjoint v. For V in Z_{app} the morphism $V \to v(V)$ is a closed immersion identifying V with $u(v(V)) = v(V) \times_X Z$ and every point of $v(V)$ specializes to a point of V. The functor v is cocontinuous and sends coverings to coverings.

Proof. The existence of the adjoint follows immediately from Lemma [original label: `lemma-lift-ind-etale`] and the definitions. It is clear that u is continuous from the definition of coverings in X_{app}.

Write $X = \operatorname{Spec}(A)$ and $Z = \operatorname{Spec}(A/I)$. Let $V = \operatorname{Spec}(\overline{C})$ be an object of Z_{app} and let $v(V) = \operatorname{Spec}(C)$. We have seen in the statement of Lemma [original label: `lemma-lift-ind-etale`] that V equals $v(V) \times_X Z = \operatorname{Spec}(C/IC)$. Any $g \in C$ which maps to an invertible element of $C/IC = \overline{C}$ is invertible in C. Namely, we have the A-algebra maps $C \to C_g \to C/IC$ and by adjointness we obtain an C-algebra map $C_g \to C$. Thus every point of $v(V)$ specializes to a point of V.

Suppose that $\{V_i \to V\}$ is a covering in Z_{app}. Then $\{v(V_i) \to v(V)\}$ is a finite family of morphisms of Z_{app} such that every point of $V \subset v(V)$ is in the image of one of the maps $v(V_i) \to v(V)$. As the morphisms $v(V_i) \to v(V)$ are flat (since they are weakly étale) we conclude that $\{v(V_i) \to v(V)\}$ is jointly surjective. This proves that v sends coverings to coverings.

Let V be an object of Z_{app} and let $\{U_i \to v(V)\}$ be a covering in X_{app}. Then we see that $\{u(U_i) \to u(v(V)) = V\}$ is a covering of Z_{app}. By adjointness we obtain morphisms $v(u(U_i)) \to U_i$. Thus the family $\{v(u(U_i)) \to v(V)\}$ refines the given covering and we conclude that v is cocontinuous. □

Lemma 7.2. Let $Z \to X$ be a closed immersion morphism of affine schemes. The corresponding morphism of topoi $i = i_{pro\text{-}étale}$ is equal to the morphism of topoi associated to the fully faithful cocontinuous functor $v : Z_{app} \to X_{app}$ of Lemma 7.1. It follows that

1. $i^{-1}\mathcal{F}$ is the sheaf associated to the presheaf $V \mapsto \mathcal{F}(v(V))$,
2. for a weakly contractible object V of Z_{app} we have $i^{-1}\mathcal{F}(V) = \mathcal{F}(v(V))$,
3. $i^{-1} : \mathrm{Sh}(X_{pro\text{-}étale}) \to \mathrm{Sh}(Z_{pro\text{-}étale})$ has a left adjoint $i_!^{Sh}$,
4. $i^{-1} : \mathrm{Ab}(X_{pro\text{-}étale}) \to \mathrm{Ab}(Z_{pro\text{-}étale})$ has a left adjoint $i_!$,
5. $\mathrm{id} \to i^{-1}i_!^{Sh}$, $\mathrm{id} \to i^{-1}i_!$, and $i^{-1}i_* \to \mathrm{id}$ are isomorphisms, and
6. i_*, $i_!^{Sh}$ and $i_!$ are fully faithful.

Proof. By Lemma [original label: `lemma-affine-alternative`] we may describe $i_{pro\text{-}étale}$ in terms of the morphism of sites $u : X_{app} \to Z_{app}$, $V \mapsto V \times_X Z$. The first statement of the lemma follows from Sites, Lemmas [original label: `sites-lemma-have-functor-other-way`] and [original label: `sites-lemma-have-functor-other-way-morphism`] (but with the roles of u and v reversed).

Proof of (1). By the description of i as the morphism of topoi associated to v this holds by the construction, see Sites, Lemma [original label: `sites-lemma-cocontinuous-morphism-topoi`].

Proof of (2). Since the functor v sends coverings to coverings by Lemma 7.1 we see that the presheaf $\mathcal{G} : V \mapsto \mathcal{F}(v(V))$ is a separated presheaf (Sites, Definition [original label: `sites-definition-separated`]). Hence the sheafification of $\mathcal{G}$ is $\mathcal{G}^+$, see Sites, Theorem [original label: `sites-theorem-plus`]. Next, let V be a weakly contractible object of Z_{app}. Let $\mathcal{V} = \{V_i \to V\}_{i=1,\ldots,n}$ be any covering in Z_{app}. Set $\mathcal{V}' = \{\coprod V_i \to V\}$. Since v commutes with finite disjoint unions (as a left adjoint or by the construction) and since $\mathcal{F}$ sends finite disjoint unions into products, we see that

$$H^0(\mathcal{V}, \mathcal{G}) = H^0(\mathcal{V}', \mathcal{G})$$

(notation as in Sites, Section [original label: `sites-section-sheafification`]; compare with Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-cech-complex`]). Thus we may assume the covering is given by a single morphism, like so $\{V' \to V\}$. Since V is weakly contractible, this covering can be refined by the trivial covering $\{V \to V\}$. It therefore follows that the value of $\mathcal{G}^+ = i^{-1}\mathcal{F}$ on V is simply $\mathcal{F}(v(V))$ and (2) is proved.

Proof of (3). Every object of Z_{app} has a covering by weakly contractible objects (Lemma [original label: `lemma-proetale-enough-w-contractible`]). By the above we see that we would have $i_!^{Sh}h_V = h_{v(V)}$ for V weakly contractible if $i_!^{Sh}$ existed. The existence of $i_!^{Sh}$ then follows from Sites, Lemma [original label: `sites-lemma-existence-lower-shriek`].

Proof of (4). Existence of $i_!$ follows in the same way by setting $i_!\mathrm{Z}_V = \mathrm{Z}_{v(V)}$ for V weakly contractible in Z_{app}, using similar for direct sums, and applying Homology, Lemma [original label: `homology-lemma-partially-defined-adjoint`]. Details omitted.

Proof of (5). Let V be a contractible object of Z_{app}. Then $i^{-1}i_!^{Sh}h_V = i^{-1}h_{v(V)} = h_{u(v(V))} = h_V$. (It is a general fact that $i^{-1}h_U = h_{u(U)}$.) Since the sheaves h_V for V contractible generate $\mathrm{Sh}(Z_{app})$ (Sites, Lemma [original label: `sites-lemma-sheaf-coequalizer-representable`]) we conclude $\mathrm{id} \to i^{-1}i_!^{Sh}$ is an isomorphism. Similarly for the map $\mathrm{id} \to i^{-1}i_!$. Then $(i^{-1}i_*\mathcal{H})(V) = i_*\mathcal{H}(v(V)) = \mathcal{H}(u(v(V))) = \mathcal{H}(V)$ and we find that $i^{-1}i_* \to \mathrm{id}$ is an isomorphism.

The fully faithfulness statements of (6) now follow from Categories, Lemma [original label: `categories-lemma-adjoint-fully-faithful`]. □

Lemma 7.3. Let $i : Z \to X$ be a closed immersion of schemes. Then

1. $i_{pro\text{-}étale}^{-1}$ commutes with limits,

2. $i_{pro\text{-}étale,*}$ is fully faithful, and

3. $i_{pro\text{-}étale}^{-1} i_{pro\text{-}étale,*} \cong \mathrm{id}_{\mathrm{Sh}(Z_{pro\text{-}étale})}$.

Proof. Assertions (2) and (3) are equivalent by Sites, Lemma [original label: `sites-lemma-exactness-properties`]. Parts (1) and (3) are (Zariski) local on X, hence we may assume that X is affine. In this case the result follows from Lemma 7.2. □

Lemma 7.4. Let $i : Z \to X$ be an integral universally injective and surjective morphism of schemes. Then $i_{pro\text{-}étale,*}$ and $i_{pro\text{-}étale}^{-1}$ are quasi-inverse equivalences of categories of pro-étale topoi.

Proof. There is an immediate reduction to the case that X is affine. Then Z is affine too. Set $A = \mathcal{O}(X)$ and $B = \mathcal{O}(Z)$. Then the categories of étale algebras over A and B are equivalent, see Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-topological-invariance`] and Remark [original label: `etale-cohomology-remark-affine-inside-equivalence`]. Thus the categories of ind-étale algebras over A and B are equivalent. In other words the categories X_{app} and Z_{app} of Lemma [original label: `lemma-affine-alternative`] are equivalent. We omit the verification that this equivalence sends coverings to coverings and vice versa. Thus the result as Lemma [original label: `lemma-affine-alternative`] tells us the pro-étale topos is the topos of sheaves on X_{app}. □

Lemma 7.5. Let $i : Z \to X$ be a closed immersion of schemes. Let $U \to X$ be an object of $X_{pro\text{-}étale}$ such that

1. U is affine and weakly contractible, and

2. every point of U specializes to a point of $U \times_X Z$.

Then $i_{pro\text{-}étale}^{-1}\mathcal{F}(U \times_X Z) = \mathcal{F}(U)$ for all abelian sheaves on $X_{pro\text{-}étale}$.

Proof. Since pullback commutes with restriction, we may replace X by U. Thus we may assume that X is affine and weakly contractible and that every point of X specializes to a point of Z. By Lemma 7.2 part (1) it suffices to show that $v(Z) = X$ in this case. Thus we have to show: If A is a w-contractible ring, $I \subset A$ an ideal contained in the Jacobson radical of A and $A \to B \to A/I$ is a factorization with $A \to B$ ind-étale, then there is a unique retraction $B \to A$ compatible with maps to A/I. Observe that $B/IB = A/I \times R$ as A/I-algebras. After replacing B by a localization we may assume $B/IB = A/I$. Note that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is surjective as the image contains $V(I)$ and hence all closed points and is closed under specialization. Since A is w-contractible there is a retraction $B \to A$. Since $B/IB = A/I$ this retraction is compatible with the map to A/I. We omit the proof of uniqueness (hint: use that A and B have isomorphic local rings at maximal ideals of A). □

Lemma 7.6. Let $i : Z \to X$ be a closed immersion of schemes. If $X \setminus i(Z)$ is a retrocompact open of X, then $i_{pro\text{-}étale,*}$ is exact.

Proof. The question is local on X hence we may assume X is affine. Say $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(A/I)$. There exist $f_1, \ldots, f_r \in I$ such that $Z = V(f_1, \ldots, f_r)$ set theoretically, see Algebra, Lemma [original label: `algebra-lemma-qc-open`]. By Lemma 7.4 we may assume that $Z = \mathrm{Spec}(A/(f_1, \ldots, f_r))$. In this case the functor $i_{pro\text{-}étale,*}$ is exact by Lemma 6.1. □

8 Extension by zero

The general material in Modules on Sites, Section [original label: `sites-modules-section-localize`] allows us to make the following definition.

Definition 8.1. Let $j : U \to X$ be a weakly étale morphism of schemes.

1. The restriction functor $j^{-1} : \mathrm{Sh}(X_{pro\text{-}étale}) \to \mathrm{Sh}(U_{pro\text{-}étale})$ has a left adjoint $j_!^{Sh} : \mathrm{Sh}(X_{pro\text{-}étale}) \to \mathrm{Sh}(U_{pro\text{-}étale})$.

2. The restriction functor $j^{-1} : \mathrm{Ab}(X_{pro\text{-}étale}) \to \mathrm{Ab}(U_{pro\text{-}étale})$ has a left adjoint which is denoted $j_! : \mathrm{Ab}(U_{pro\text{-}étale}) \to \mathrm{Ab}(X_{pro\text{-}étale})$ and called extension by zero.

3. Let Λ be a ring. The functor $j^{-1} : \mathrm{Mod}(X_{pro\text{-}étale}, \Lambda) \to \mathrm{Mod}(U_{pro\text{-}étale}, \Lambda)$ has a left adjoint $j_! : \mathrm{Mod}(U_{pro\text{-}étale}, \Lambda) \to \mathrm{Mod}(X_{pro\text{-}étale}, \Lambda)$ and called extension by zero.

As usual we compare this to what happens in the étale case.

Lemma 8.2. Let $j : U \to X$ be an étale morphism of schemes. Let $\mathcal{G}$ be an abelian sheaf on $U_{étale}$. Then $\epsilon^{-1} j_! \mathcal{G} = j_! \epsilon^{-1} \mathcal{G}$ as sheaves on $X_{pro\text{-}étale}$.

Proof. This is true because both functors are left adjoint to $j_{pro\text{-}étale}^{-1}\epsilon_* = \epsilon_* j_{étale}^{-1}$. The equality holds by the discussion in Section 5. □

Lemma 8.3. Let $j : U \to X$ be a weakly étale morphism of schemes. Let $i : Z \to X$ be a closed immersion such that $U \times_X Z = \emptyset$. Let $V \to X$ be an affine object of $X_{pro\text{-}étale}$ such that every point of V specializes to a point of $V_Z = Z \times_X V$. Then $j_!\mathcal{F}(V) = 0$ for all abelian sheaves on $U_{pro\text{-}étale}$.

Proof. Let $\{V_i \to V\}$ be a pro-étale covering. The lemma follows if we can refine this covering to a covering where the members have no morphisms into U over X (see construction of $j_!$ in Modules on Sites, Section [original label: `sites-modules-section-localize`]). First refine the covering to get a finite covering with V_i affine. For each i let $V_i = \mathrm{Spec}(A_i)$ and let $Z_i \subset V_i$ be the inverse image of Z. Set $W_i = \mathrm{Spec}(A_{i,Z_i}^\sim)$ with notation as in Lemma [original label: `lemma-localization`]. Then $\coprod W_i \to V$ is weakly étale and the image contains all points of V_Z. Hence the image contains all points of V by our assumption on specializations. Thus $\{W_i \to V\}$ is a pro-étale covering refining the given one. But each point in W_i specializes to a point lying over Z, hence there are no morphisms $W_i \to U$ over X. □

Lemma 8.4. Let $j : U \to X$ be an open immersion of schemes. Then $\mathrm{id} \cong j^{-1}j_!$ and $j^{-1}j_* \cong \mathrm{id}$ and the functors $j_!$ and j_* are fully faithful.

Proof. See Modules on Sites, Lemma [original label: `sites-modules-lemma-restrict-back`] (and Sites, Lemma [original label: `sites-lemma-restrict-back`] for the case of sheaves of sets) and Categories, Lemma [original label: `categories-lemma-adjoint-fully-faithful`]. □

Here is the relationship between extension by zero and restriction to the complementary closed subscheme.

Lemma 8.5. Let X be a scheme. Let $Z \subset X$ be a closed subscheme and let $U \subset X$ be the complement. Denote $i : Z \to X$ and $j : U \to X$ the inclusion morphisms. Assume that j is a quasi-compact morphism. For every abelian sheaf on $X_{pro\text{-}étale}$ there is a canonical short exact sequence

$$0 \to j_!j^{-1}\mathcal{F} \to \mathcal{F} \to i_*i^{-1}\mathcal{F} \to 0$$

on $X_{pro\text{-}étale}$ where all the functors are for the pro-étale topology.

Proof. We obtain the maps by the adjointness properties of the functors involved. It suffices to show that $X_{pro\text{-}étale}$ has enough objects (Sites, Definition [original label: `sites-definition-w-contractible`]) on which the sequence evaluates to a short exact sequence. Let $V = \mathrm{Spec}(A)$ be an affine object of $X_{pro\text{-}étale}$ such that A is w-contractible (there are enough objects of this type). Then $V \times_X Z$ is cut out by an ideal $I \subset A$. The assumption that j is quasi-compact implies there exist $f_1, \ldots, f_r \in I$ such that $V(I) = V(f_1, \ldots, f_r)$. We obtain a faithfully flat, ind-Zariski ring map

$$A \longrightarrow A_{f_1} \times \ldots \times A_{f_r} \times A_{V(I)}^\sim$$

with $A_{V(I)}^\sim$ as in Lemma [original label: `lemma-localization`]. Since $V_i = \mathrm{Spec}(A_{f_i}) \to X$ factors through U we have

$$j_!j^{-1}\mathcal{F}(V_i) = \mathcal{F}(V_i) \quad \text{and} \quad i_*i^{-1}\mathcal{F}(V_i) = 0$$

On the other hand, for the scheme $V^\sim = \mathrm{Spec}(A_{V(I)}^\sim)$ we have

$$j_!j^{-1}\mathcal{F}(V^\sim) = 0 \quad \text{and} \quad \mathcal{F}(V^\sim) = i_*i^{-1}\mathcal{F}(V^\sim)$$

the first equality by Lemma 8.3 and the second by Lemmas 7.5 and [original label: `lemma-localization-w-contractible`]. Thus the sequence evaluates to an exact sequence on $\mathrm{Spec}(A_{f_1} \times \ldots \times A_{f_r} \times A_{V(I)}^\sim)$ and the lemma is proved. □

Lemma 8.6. Let $j : U \to X$ be a quasi-compact open immersion morphism of schemes. The functor $j_! : \mathrm{Ab}(U_{pro\text{-}étale}) \to \mathrm{Ab}(X_{pro\text{-}étale})$ commutes with limits.

Proof. Since $j_!$ is exact it suffices to show that $j_!$ commutes with products. The question is local on X, hence we may assume X affine. Let $\mathcal{G}$ be an abelian sheaf on $U_{pro\text{-}étale}$. We have $j^{-1}j_*\mathcal{G} = \mathcal{G}$. Hence applying the exact sequence of Lemma 8.5 we get

$$0 \to j_!\mathcal{G} \to j_*\mathcal{G} \to i_*i^{-1}j_*\mathcal{G} \to 0$$

where $i : Z \to X$ is the inclusion of the reduced induced scheme structure on the complement $Z = X \setminus U$. The functors j_* and i_* commute with products as right adjoints. The functor i^{-1} commutes with products by Lemma 7.3. Hence $j_!$ does because on the pro-étale site products are exact (Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`]). □

9 Constructible sheaves on the pro-étale site

We stick to constructible sheaves of Λ-modules for a Noetherian ring. In the future we intend to discuss constructible sheaves of sets, groups, etc.

Definition 9.1. Let X be a scheme. Let Λ be a Noetherian ring. A sheaf of Λ-modules on $X_{pro\text{-}étale}$ is constructible if for every affine open $U \subset X$ there exists a finite decomposition of U into constructible locally closed subschemes $U = \coprod_i U_i$ such that $\mathcal{F}|_{U_i}$ is of finite type and locally constant for all i.

Again this does not give anything "new".

Lemma 9.2. Let X be a scheme. Let Λ be a Noetherian ring. The functor ϵ^{-1} defines an equivalence of categories

$$\left\{\begin{array}{l}\text{constructible sheaves of} \\ \Lambda\text{-modules on } X_{étale}\end{array}\right\} \longleftrightarrow \left\{\begin{array}{l}\text{constructible sheaves of} \\ \Lambda\text{-modules on } X_{pro\text{-}étale}\end{array}\right\}$$

between constructible sheaves of Λ-modules on $X_{étale}$ and constructible sheaves of Λ-modules on $X_{pro\text{-}étale}$.

Proof. By Lemma 1.2 the functor ϵ^{-1} is fully faithful and commutes with pullback (restriction) to the strata. Hence ϵ^{-1} of a constructible étale sheaf is a constructible pro-étale sheaf. To finish the proof let $\mathcal{F}$ be a constructible sheaf of Λ-modules on $X_{pro\text{-}étale}$ as in Definition 9.1. There is a canonical map

$$\epsilon^{-1}\epsilon_*\mathcal{F} \longrightarrow \mathcal{F}$$

We will show this map is an isomorphism. This will prove that $\mathcal{F}$ is in the essential image of ϵ^{-1} and finish the proof (details omitted).

Since it suffices to prove this locally on X we may assume X is quasi-compact and quasi-separated and that we have a finite partition $X = \coprod_{i=1,\ldots,n} X_i$ by constructible locally closed strata such that $\mathcal{F}|_{X_i}$ is locally constant of finite type. We will use induction on n. The base case $n = 1$ follows from Lemma 1.9. Take a point $x \in X$; then $x \in X_k$ for some k. It suffices to show the displayed map is an isomorphism in an open neighbourhood of x. Hence we may assume X_k is closed in X. Set $Z = X_k$ and denote $U \subset X$ the complement of Z. Observe that the induction hypothesis applies to the restriction of $\mathcal{F}$ to Z and to U. By Lemma 8.5 we have a short exact sequence

$$0 \to j_!j^{-1}\mathcal{F} \to \mathcal{F} \to i_*i^{-1}\mathcal{F} \to 0$$

on $X_{pro\text{-}étale}$. Functoriality gives a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \epsilon^{-1}\epsilon_*j_!j^{-1}\mathcal{F} & \longrightarrow & \epsilon^{-1}\epsilon_*\mathcal{F} & \longrightarrow & \epsilon^{-1}\epsilon_*i_*i^{-1}\mathcal{F} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & j_!j^{-1}\mathcal{F} & \longrightarrow & \mathcal{F} & \longrightarrow & i_*i^{-1}\mathcal{F} & \longrightarrow & 0 \end{array}$$

By induction we know that on the one hand $\epsilon^{-1}\epsilon_*i^{-1}\mathcal{F} \to i^{-1}\mathcal{F}$ and $\epsilon^{-1}\epsilon_*j^{-1}\mathcal{F} \to j^{-1}\mathcal{F}$ are isomorphisms and on the other that $i^{-1}\mathcal{F} = \epsilon^{-1}\mathcal{A}$ and $j^{-1}\mathcal{F} = \epsilon^{-1}\mathcal{B}$ for some constructible sheaves of Λ-modules $\mathcal{A}$ on $Z_{étale}$ and $\mathcal{B}$ on $U_{étale}$. Then

$$\epsilon^{-1}\epsilon_*j_!j^{-1}\mathcal{F} = \epsilon^{-1}\epsilon_*j_!\epsilon^{-1}\mathcal{B} = \epsilon^{-1}\epsilon_*\epsilon^{-1}j_!\mathcal{B} = \epsilon^{-1}j_!\mathcal{B} = j_!\epsilon^{-1}\mathcal{B} = j_!j^{-1}\mathcal{F}$$

the second equality by Lemma 8.2, the third equality by Lemma 1.2, and the fourth equality by Lemma 8.2 again. Similarly, we have

$$\epsilon^{-1}\epsilon_*i_*i^{-1}\mathcal{F} = \epsilon^{-1}\epsilon_*i_*\epsilon^{-1}\mathcal{A} = \epsilon^{-1}\epsilon_*\epsilon^{-1}i_*\mathcal{A} = \epsilon^{-1}i_*\mathcal{A} = i_*\epsilon^{-1}\mathcal{A} = i_*i^{-1}\mathcal{F}$$

this time using Lemma 5.1. By the five lemma we conclude the vertical map in the middle of the big diagram is an isomorphism. □

Lemma 9.3. Let X be a scheme. Let Λ be a Noetherian ring. The category of constructible sheaves of Λ-modules on $X_{pro\text{-}étale}$ is a weak Serre subcategory of $\mathrm{Mod}(X_{pro\text{-}étale}, \Lambda)$.

Proof. This is a formal consequence of Lemmas 9.2 and 1.8 and the result for the étale site (Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-constructible-abelian`]). □

Lemma 9.4. Let X be a scheme. Let Λ be a Noetherian ring. Let $D_c(X_{étale}, \Lambda)$, resp. $D_c(X_{pro\text{-}étale}, \Lambda)$ be the full subcategory of $D(X_{étale}, \Lambda)$, resp. $D(X_{pro\text{-}étale}, \Lambda)$ consisting of those complexes whose cohomology sheaves are constructible sheaves of Λ-modules. Then

$$\epsilon^{-1} : D_c^+(X_{étale}, \Lambda) \longrightarrow D_c^+(X_{pro\text{-}étale}, \Lambda)$$

is an equivalence of categories.

Proof. The categories $D_c(X_{étale}, \Lambda)$ and $D_c(X_{pro\text{-}étale}, \Lambda)$ are strictly full, saturated, triangulated subcategories of $D(X_{étale}, \Lambda)$ and $D(X_{pro\text{-}étale}, \Lambda)$ by Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-constructible-abelian`] and Lemma 9.3 and Derived Categories, Section [original label: `derived-section-triangulated-sub`]. The statement of the lemma follows by combining Lemmas 1.8 and 9.2. □

Lemma 9.5. Let X be a scheme. Let Λ be a Noetherian ring. Let $K, L \in D_c^-(X_{pro\text{-}étale}, \Lambda)$. Then $K \otimes^{\mathrm{L}}_{\Lambda} L$ is in $D_c^-(X_{pro\text{-}étale}, \Lambda)$.

Proof. Note that $H^i(K \otimes^{\mathrm{L}}_{\Lambda} L)$ is the same as $H^i(\tau_{\geq i-1}K \otimes^{\mathrm{L}}_{\Lambda} \tau_{\geq i-1}L)$. Thus we may assume K and L are bounded. In this case we can apply Lemma 9.4 to reduce to the case of the étale site, see Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-tensor-c`]. □

Lemma 9.6. Let X be a scheme. Let Λ be a Noetherian ring. Let $I \subset \Lambda$ be an ideal. Let K be an object of $D(X_{pro\text{-}étale}, \Lambda)$. Set $K_n = K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. If K_1 is in $D_c^-(X_{pro\text{-}étale}, \Lambda/I)$, then K_n is in $D_c^-(X_{pro\text{-}étale}, \Lambda/I^n)$ for all n.

Proof. Consider the distinguished triangles

$$K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}} \to K_{n+1} \to K_n \to K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}}[1]$$

and the isomorphisms

$$K \otimes^{\mathrm{L}}_{\Lambda} \underline{I^n/I^{n+1}} = K_1 \otimes^{\mathrm{L}}_{\Lambda/I} \underline{I^n/I^{n+1}}$$

By Lemma 9.5 we see that this tensor product has constructible cohomology sheaves (and vanishing when K_1 has vanishing cohomology). Hence by induction on n using Lemma 9.3 we see that each K_n has constructible cohomology sheaves. □

10 Constructible adic sheaves

In this section we define the notion of a constructible Λ-sheaf as well as some variants.

Definition 10.1. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let X be a scheme. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{pro\text{-}étale}$.

1. We say $\mathcal{F}$ is a constructible Λ-sheaf if $\mathcal{F} = \lim \mathcal{F}/I^n\mathcal{F}$ and each $\mathcal{F}/I^n\mathcal{F}$ is a constructible sheaf of Λ/I^n-modules.
2. If $\mathcal{F}$ is a constructible Λ-sheaf, then we say $\mathcal{F}$ is lisse if each $\mathcal{F}/I^n\mathcal{F}$ is locally constant.
3. We say $\mathcal{F}$ is adic lisse[2] if there exists a I-adically complete Λ-module M with M/IM finite such that $\mathcal{F}$ is locally isomorphic to
$$\underline{M}^\wedge = \lim \underline{M/I^nM}.$$
4. We say $\mathcal{F}$ is adic constructible[3] if for every affine open $U \subset X$ there exists a decomposition $U = \coprod U_i$ into constructible locally closed subschemes such that $\mathcal{F}|_{U_i}$ is adic lisse.

The definition of a constructible Λ-sheaf is equivalent to the one in [2, Exposé VI, Definition 1.1.1] when $\Lambda = \mathrm{Z}_\ell$ and $I = (\ell)$. It is clear that we have the implications

$$\begin{array}{ccc} \text{lisse adic} & \Longrightarrow & \text{adic constructible} \\ \Downarrow & & \Downarrow \\ \text{lisse constructible } \Lambda\text{-sheaf} & \Longrightarrow & \text{constructible } \Lambda\text{-sheaf} \end{array}$$

The vertical arrows can be inverted in some cases (see Lemmas 10.2 and 10.5). In general neither the category of adic constructible sheaves nor the category of constructible Λ-sheaves is closed under kernels and cokernels.

Namely, let X be an affine scheme whose underlying topological space $|X|$ is homeomorphic to $\Lambda = \mathrm{Z}_\ell$, see Example [original label: `example-construct-space`]. Denote $f : |X| \to \mathrm{Z}_\ell = \Lambda$ a homeomorphism. We can think of f as a section of $\underline{\Lambda}^\wedge$ over X and multiplication by f then defines a two term complex

$$\underline{\Lambda}^\wedge \xrightarrow{f} \underline{\Lambda}^\wedge$$

on $X_{pro\text{-}étale}$. The sheaf $\underline{\Lambda}^\wedge$ is adic lisse. However, the cokernel of the map above, is not adic constructible, as the isomorphism type of the stalks of this cokernel attains infinitely many values: $\mathrm{Z}/\ell^n\mathrm{Z}$ and Z_ℓ. The cokernel is a constructible Z_ℓ-sheaf. However, the kernel is not even a constructible Z_ℓ-sheaf as it is zero a non-quasi-compact open but not zero.

[2] This may be nonstandard notation.
[3] This may be nonstandard notation.

Lemma 10.2. Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a constructible Λ-sheaf on $X_{pro\text{-}étale}$. Then there exists a finite partition $X = \coprod X_i$ by locally closed subschemes such that the restriction $\mathcal{F}|_{X_i}$ is lisse.

Proof. Let $R = \bigoplus I^n/I^{n+1}$. Observe that R is a Noetherian ring. Since each of the sheaves $\mathcal{F}/I^n\mathcal{F}$ is a constructible sheaf of $\Lambda/I^n\Lambda$-modules also $I^n\mathcal{F}/I^{n+1}\mathcal{F}$ is a constructible sheaf of Λ/I-modules and hence the pullback of a constructible sheaf $\mathcal{G}_n$ on $X_{étale}$ by Lemma 9.2. Set $\mathcal{G} = \bigoplus \mathcal{G}_n$. This is a sheaf of R-modules on $X_{étale}$ and the map

$$\mathcal{G}_0 \otimes_{\Lambda/I} \underline{R} \longrightarrow \mathcal{G}$$

is surjective because the maps

$$\mathcal{F}/I\mathcal{F} \otimes \underline{I^n/I^{n+1}} \to I^n\mathcal{F}/I^{n+1}\mathcal{F}$$

are surjective. Hence $\mathcal{G}$ is a constructible sheaf of R-modules by Étale Cohomology, Proposition [original label: `etale-cohomology-proposition-constructible-over-noetherian`]. Choose a partition $X = \coprod X_i$ such that $\mathcal{G}|_{X_i}$ is a locally constant sheaf of R-modules of finite type (Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-constructible-quasi-compact-quasi-separated`]). We claim this is a partition as in the lemma. Namely, replacing X by X_i we may assume $\mathcal{G}$ is locally constant. It follows that each of the sheaves $I^n\mathcal{F}/I^{n+1}\mathcal{F}$ is locally constant. Using the short exact sequences

$$0 \to I^n\mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^n\mathcal{F} \to 0$$

induction and Modules on Sites, Lemma [original label: `sites-modules-lemma-kernel-finite-locally-constant`] the lemma follows. □

Lemma 10.3. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{pro\text{-}étale}$ such that

1. $\mathcal{F} = \lim \mathcal{F}/I^n\mathcal{F}$,
2. $\mathcal{F}/I^n\mathcal{F}$ is a constant sheaf of Λ/I^n-modules,
3. $\mathcal{F}/I\mathcal{F}$ is of finite type.

Then $\mathcal{F} \cong \underline{M}^\wedge$ where M is a finite $\Lambda^\wedge$-module.

Proof. Pick a Λ/I^n-module M_n such that $\mathcal{F}/I^n\mathcal{F} \cong \underline{M_n}$. Since we have the surjections $\mathcal{F}/I^{n+1}\mathcal{F} \to \mathcal{F}/I^n\mathcal{F}$ we conclude that there exist surjections $M_{n+1} \to M_n$ inducing isomorphisms $M_{n+1}/I^nM_{n+1} \to M_n$. Fix a choice of such surjections and set $M = \lim M_n$. Then M is an I-adically complete Λ-module with $M/I^nM = M_n$, see Algebra, Lemma [original label: `algebra-lemma-limit-complete`]. Since M_1 is a finite type Λ-module (Modules on Sites, Lemma [original label: `sites-modules-lemma-locally-constant-finite-type`]) we see that M is a finite $\Lambda^\wedge$-module. Consider the sheaves

$$\mathcal{I}_n = \mathrm{Isom}(\underline{M_n}, \mathcal{F}/I^n\mathcal{F})$$

on $X_{pro\text{-}étale}$. Modding out by I^n defines a transition map

$$\mathcal{I}_{n+1} \longrightarrow \mathcal{I}_n$$

By our choice of M_n the sheaf $\mathcal{I}_n$ is a torsor under

$$\mathrm{Isom}(\underline{M_n}, \underline{M_n}) = \underline{\mathrm{Isom}_\Lambda(M_n, M_n)}$$

(Modules on Sites, Lemma [original label: `sites-modules-lemma-locally-constant`]) since $\mathcal{F}/I^n\mathcal{F}$ is (étale) locally isomorphic to $\underline{M_n}$. It follows from More on Algebra, Lemma [original label: `more-algebra-lemma-hom-systems-ML`] that the system of sheaves $(\mathcal{I}_n)$ is Mittag-Leffler. For each n let $\mathcal{I}'_n \subset \mathcal{I}_n$ be the image of $\mathcal{I}_N \to \mathcal{I}_n$ for all $N \gg n$. Then

$$\ldots \to \mathcal{I}'_3 \to \mathcal{I}'_2 \to \mathcal{I}'_1 \to *$$

is a sequence of sheaves of sets on $X_{pro\text{-}étale}$ with surjective transition maps. Since $*(X)$ is a singleton (not empty) and since evaluating at X transforms surjective maps of sheaves of sets into surjections of sets, we can pick $s \in \lim \mathcal{I}'_n(X)$. The sections define isomorphisms $\underline{M}^\wedge \to \lim \mathcal{F}/I^n\mathcal{F} = \mathcal{F}$ and the proof is done. □

Lemma 10.4. Let X be a connected scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. If $\mathcal{F}$ is a lisse constructible Λ-sheaf on $X_{pro\text{-}étale}$, then $\mathcal{F}$ is adic lisse.

Proof. By Lemma 1.9 we have $\mathcal{F}/I^n\mathcal{F} = \epsilon^{-1}\mathcal{G}_n$ for some locally constant sheaf $\mathcal{G}_n$ of Λ/I^n-modules. By Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-connected-locally-constant`] there exists a finite Λ/I^n-module M_n such that $\mathcal{G}_n$ is locally isomorphic to $\underline{M_n}$. Choose a covering $\{W_t \to X\}_{t\in T}$ with each W_t affine and weakly contractible. Then $\mathcal{F}|_{W_t}$ satisfies the assumptions of Lemma 10.3 and hence $\mathcal{F}|_{W_t} \cong \underline{N_t}^\wedge$ for some finite $\Lambda^\wedge$-module N_t. Note that $N_t/I^nN_t \cong M_n$ for all t and n. Hence $N_t \cong N_{t'}$ for all $t, t' \in T$, see More on Algebra, Lemma [original label: `more-algebra-lemma-isomorphic-completions`]. This proves that $\mathcal{F}$ is adic lisse. □

Lemma 10.5. Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a constructible Λ-sheaf on $X_{pro\text{-}étale}$. Then $\mathcal{F}$ is adic constructible.

Proof. This is a consequence of Lemmas 10.2 and 10.4, the fact that a Noetherian scheme is locally connected (Topology, Lemma [original label: `topology-lemma-locally-Noetherian-locally-connected`]), and the definitions. □

It will be useful to identify the constructible Λ-sheaves inside the category of derived complete sheaves of Λ-modules. It turns out that the naive analogue of More on Algebra, Lemma [original label: `more-algebra-lemma-derived-complete-finite`] is wrong in this setting. However, here is the analogue of More on Algebra, Lemma [original label: `more-algebra-lemma-derived-complete-zero`].

Lemma 10.6. Let X be a scheme. Let Λ be a ring and let $I \subset \Lambda$ be a finitely generated ideal. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{pro\text{-}étale}$. If $\mathcal{F}$ is derived complete and $\mathcal{F}/I\mathcal{F} = 0$, then $\mathcal{F} = 0$.

Proof. Assume that $\mathcal{F}/I\mathcal{F}$ is zero. Let $I = (f_1, \ldots, f_r)$. Let $i < r$ be the largest integer such that $\mathcal{G} = \mathcal{F}/(f_1, \ldots, f_i)\mathcal{F}$ is nonzero. If i does not exist, then $\mathcal{F} = 0$ which is what we want to show. Then $\mathcal{G}$ is derived complete as a cokernel of a map between derived complete modules, see Proposition 3.1. By our choice of i we have that $f_{i+1} : \mathcal{G} \to \mathcal{G}$ is surjective. Hence

$$\lim(\ldots \to \mathcal{G} \xrightarrow{f_{i+1}} \mathcal{G} \xrightarrow{f_{i+1}} \mathcal{G})$$

is nonzero, contradicting the derived completeness of $\mathcal{G}$. □

Lemma 10.7. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $\mathcal{F}$ be a derived complete sheaf of Λ-modules on $X_{pro\text{-}étale}$ with $\mathcal{F}/I\mathcal{F}$ a locally constant sheaf of Λ/I-modules of finite type. Then there exists an integer t and a surjective map

$$(\underline{\Lambda}^\wedge)^{\oplus t} \to \mathcal{F}$$

Proof. Since X is weakly contractible, there exists a finite disjoint open covering $X = \coprod U_i$ such that $\mathcal{F}/I\mathcal{F}|_{U_i}$ is isomorphic to the constant sheaf associated to a finite Λ/I-module M_i. Choose finitely many generators m_{ij} of M_i. We can find sections $s_{ij} \in \mathcal{F}(X)$ restricting to m_{ij} viewed as a section of $\mathcal{F}/I\mathcal{F}$ over U_i. Let t be the total number of s_{ij}. Then we obtain a map

$$\alpha : \underline{\Lambda}^{\oplus t} \longrightarrow \mathcal{F}$$

which is surjective modulo I by construction. By Lemma 2.1 the derived completion of $\underline{\Lambda}^{\oplus t}$ is the sheaf $(\underline{\Lambda}^\wedge)^{\oplus t}$. Since $\mathcal{F}$ is derived complete we see that α factors through a map

$$\alpha^\wedge : (\underline{\Lambda}^\wedge)^{\oplus t} \longrightarrow \mathcal{F}$$

Then $\mathcal{Q} = \mathrm{Coker}(\alpha^\wedge)$ is a derived complete sheaf of Λ-modules by Proposition 3.1. By construction $\mathcal{Q}/I\mathcal{Q} = 0$. It follows from Lemma 10.6 that $\mathcal{Q} = 0$ which is what we wanted to show. □

11 A suitable derived category

Let X be a scheme. It will turn out that for many schemes X a suitable derived category of ℓ-adic sheaves can be gotten by considering the derived complete objects K of $D(X_{pro\text{-}étale}, \Lambda)$ with the property that $K \otimes^{\mathrm{L}}_\Lambda \mathrm{F}_\ell$ is bounded with constructible cohomology sheaves. Here is the general definition.

Definition 11.1. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let X be a scheme. An object K of $D(X_{pro\text{-}étale}, \Lambda)$ is called constructible if

1. K is derived complete with respect to I,
2. $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has constructible cohomology sheaves and locally has finite tor dimension.

We denote $D_{cons}(X, \Lambda)$ the full subcategory of constructible K in $D(X_{pro\text{-}étale}, \Lambda)$.

Recall that with our conventions a complex of finite tor dimension is bounded (Cohomology on Sites, Definition [original label: `sites-cohomology-definition-tor-amplitude`]). In fact, let's collect everything proved so far in a lemma.

Lemma 11.2. In the situation above suppose K is in $D_{cons}(X, \Lambda)$ and X is quasi-compact. Set $K_n = K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. There exist a, b such that

1. $K = R\lim K_n$ and $H^i(K) = 0$ for $i \notin [a, b]$,
2. each K_n has tor amplitude in $[a, b]$,
3. each K_n has constructible cohomology sheaves,
4. each $K_n = \epsilon^{-1} L_n$ for some $L_n \in D_{ctf}(X_{étale}, \Lambda/I^n)$ (Étale Cohomology, Definition [original label: *`etale-cohomology-definition-ctf`*]).

Proof. Since K is derived complete, we have $K = R\lim K_n$ by Lemma 2.1. By definition of locally having finite tor dimension, we can find a, b such that K_1 has tor amplitude in $[a, b]$. Part (2) follows from Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-bounded`]. Then (1) follows as K is derived complete by the description of limits in Cohomology on Sites, Proposition [original label: `sites-cohomology-proposition-enough-weakly-contractibles`] and the fact that $H^b(K_{n+1}) \to H^b(K_n)$ is surjective as $K_n = K_{n+1} \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. Part (3) follows from Lemma 9.6, Part (4) follows from Lemma 9.4 and the fact that L_n has finite tor dimension because K_n does (small argument omitted). □

Lemma 11.3. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let K be an object of $D_{cons}(X, \Lambda)$ such that the cohomology sheaves of $K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I}$ are locally constant. Then there exists a finite disjoint open covering $X = \coprod U_i$ and for each i a finite collection of finite projective $\Lambda^\wedge$-modules $M^a, \ldots, M^b$ such that $K|_{U_i}$ is represented by a complex

$$(\underline{M^a})^\wedge \to \ldots \to (\underline{M^b})^\wedge$$

in $D(U_{i,pro\text{-}étale}, \Lambda)$ for some maps of sheaves of Λ-modules $(\underline{M^i})^\wedge \to (\underline{M^{i+1}})^\wedge$.

Proof. We freely use the results of Lemma 11.2. Choose a, b as in that lemma. We will prove the lemma by induction on $b - a$. Let $\mathcal{F} = H^b(K)$. Note that $\mathcal{F}$ is a derived complete sheaf of Λ-modules by Proposition 3.1. Moreover $\mathcal{F}/I\mathcal{F}$ is a locally constant sheaf of Λ/I-modules of finite type. Apply Lemma 10.7 to get a surjection $\rho : (\underline{\Lambda}^\wedge)^{\oplus t} \to \mathcal{F}$.

If $a = b$, then $K = \mathcal{F}[-b]$. In this case we see that

$$\mathcal{F} \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I} = \mathcal{F}/I\mathcal{F}$$

As X is weakly contractible and $\mathcal{F}/I\mathcal{F}$ locally constant, we can find a finite disjoint union decomposition $X = \coprod U_i$ by affine opens U_i and Λ/I-modules $\overline{M}_i$ such that $\mathcal{F}/I\mathcal{F}$ restricts to $\underline{\overline{M}_i}$ on U_i. After refining the covering we may assume the map

$$\rho|_{U_i} \bmod I : \underline{\Lambda/I}^{\oplus t} \longrightarrow \underline{\overline{M}_i}$$

is equal to $\underline{\alpha_i}$ for some surjective module map $\alpha_i : \Lambda/I^{\oplus t} \to \overline{M}_i$, see Modules on Sites, Lemma [original label: `sites-modules-lemma-morphism-locally-constant`]. Note that each $\overline{M}_i$ is a finite Λ/I-module. Since $\mathcal{F}/I\mathcal{F}$ has tor amplitude in $[0, 0]$ we conclude that $\overline{M}_i$ is a flat Λ/I-module. Hence $\overline{M}_i$ is finite projective (Algebra, Lemma [original label: `algebra-lemma-finite-projective`]). Hence we can find a projector $\overline{p}_i : (\Lambda/I)^{\oplus t} \to (\Lambda/I)^{\oplus t}$ whose image maps isomorphically to $\overline{M}_i$ under the map α_i. We can lift $\overline{p}_i$ to a projector $p_i : (\Lambda^\wedge)^{\oplus t} \to (\Lambda^\wedge)^{\oplus t}$[4]. Then $M_i = \mathrm{Im}(p_i)$ is a finite I-adically complete $\Lambda^\wedge$-module with $M_i/IM_i = \overline{M}_i$. Over U_i consider the maps

$$\underline{M_i}^\wedge \to (\underline{\Lambda}^\wedge)^{\oplus t} \to \mathcal{F}|_{U_i}$$

By construction the composition induces an isomorphism modulo I. The source and target are derived complete, hence so are the cokernel $\mathcal{Q}$ and the kernel $\mathcal{K}$. We have $\mathcal{Q}/I\mathcal{Q} = 0$ by construction hence $\mathcal{Q}$ is zero by Lemma 10.6. Then

$$0 \to \mathcal{K}/I\mathcal{K} \to \underline{\overline{M}_i} \to \mathcal{F}/I\mathcal{F} \to 0$$

[4]Proof: by Algebra, Lemma [original label: `algebra-lemma-lift-idempotents-noncommutative`] we can lift $\overline{p}_i$ to a compatible system of projectors $p_{i,n} : (\Lambda/I^n)^{\oplus t} \to (\Lambda/I^n)^{\oplus t}$ and then we set $p_i = \lim p_{i,n}$ which works because $\Lambda^\wedge = \lim \Lambda/I^n$.

is exact by the vanishing of Tor_1 see at the start of this paragraph; also use that $\underline{\Lambda}^\wedge/I\overline{\Lambda}^\wedge$ by Modules on Sites, Lemma [original label: `sites-modules-lemma-completion-flat`] to see that $\underline{M_i}^\wedge/I\underline{M_i}^\wedge = \overline{M}_i$. Hence $\mathcal{K}/I\mathcal{K} = 0$ by construction and we conclude that $\mathcal{K} = 0$ as before. This proves the result in case $a = b$.

If $b > a$, then we lift the map ρ to a map

$$\tilde{\rho} : (\underline{\Lambda}^\wedge)^{\oplus t}[-b] \longrightarrow K$$

in $D(X_{pro\text{-}étale}, \Lambda)$. This is possible as we can think of K as a complex of $\underline{\Lambda}^\wedge$-modules by discussion in the introduction to Section 2 and because $X_{pro\text{-}étale}$ is weakly contractible hence there is no obstruction to lifting the elements $\rho(e_s) \in H^0(X, \mathcal{F})$ to elements of $H^b(X, K)$. Fitting $\tilde{\rho}$ into a distinguished triangle

$$(\underline{\Lambda}^\wedge)^{\oplus t}[-b] \to K \to L \to (\underline{\Lambda}^\wedge)^{\oplus t}[-b+1]$$

we see that L is an object of $D_{cons}(X, \Lambda)$ such that $L \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has tor amplitude contained in $[a, b-1]$ (details omitted). By induction we can describe L locally as stated in the lemma, say L is isomorphic to

$$(\underline{M^a})^\wedge \to \ldots \to (\underline{M^{b-1}})^\wedge$$

The map $L \to (\underline{\Lambda}^\wedge)^{\oplus t}[-b+1]$ corresponds to a map $(\underline{M^{b-1}})^\wedge \to (\underline{\Lambda}^\wedge)^{\oplus t}$ which allows us to extend the complex by one. The corresponding complex is isomorphic to K in the derived category by the properties of triangulated categories. This finishes the proof. □

Motivated by what happens for constructible Λ-sheaves we introduce the following notion.

Definition 11.4. Let X be a scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let $K \in D(X_{pro\text{-}étale}, \Lambda)$.

1. We say K is adic lisse[5] if there exists a finite complex of finite projective $\Lambda^\wedge$-modules $M^\bullet$ such that K is locally isomorphic to
$$\underline{M^a}^\wedge \to \ldots \to \underline{M^b}^\wedge$$
2. We say K is adic constructible[6] if for every affine open $U \subset X$ there exists a decomposition $U = \coprod U_i$ into constructible locally closed subschemes such that $K|_{U_i}$ is adic lisse.

The difference between the local structure obtained in Lemma 11.3 and the structure of an adic lisse complex is that the maps $\underline{M^i}^\wedge \to \underline{M^{i+1}}^\wedge$ in Lemma 11.3 need not be constant, whereas in the definition above they are required to be constant.

Lemma 11.5. Let X be a weakly contractible affine scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let K be an object of $D_{cons}(X, \Lambda)$ such that $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$ is isomorphic in $D(X_{pro\text{-}étale}, \Lambda/I^n)$ to a complex of constant sheaves of Λ/I^n-modules. Then

$$H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^n)$$

has the Mittag-Leffler condition.

Proof. Say $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$ is isomorphic to $\underline{E_n}$ for some object E_n of $D(\Lambda/I^n)$. Since $K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I}$ has finite tor dimension and has finite type cohomology sheaves we see that E_1 is perfect (see More on Algebra, Lemma [original label: `more-algebra-lemma-perfect`]). The transition maps

$$K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^{n+1}} \to K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$$

locally come from (possibly many distinct) maps of complexes $E_{n+1} \to E_n$ in $D(\Lambda/I^{n+1})$ see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-locally-constant-map`]. For each n choose one such map and observe that it induces an isomorphism $E_{n+1} \otimes^{\mathrm{L}}_{\Lambda/I^{n+1}} \Lambda/I^n \to E_n$ in $D(\Lambda/I^n)$. By More on Algebra, Lemma [original label: `more-algebra-lemma-Rlim-perfect-gives-complete`] we can find a finite complex $M^\bullet$ of finite projective $\Lambda^\wedge$-modules and isomorphisms $M^\bullet/I^nM^\bullet \to E_n$ in $D(\Lambda/I^n)$ compatible with the transition maps.

Now observe that for each finite collection of indices $n > m > k$ the triple of maps

$$H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^n) \to H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^m) \to H^0(X, K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^k)$$

[5] This may be nonstandard notation

[6] This may be nonstandard notation.

is isomorphic to

$$H^0(X, \underline{M^\bullet/I^nM^\bullet}) \to H^0(X, \underline{M^\bullet/I^mM^\bullet}) \to H^0(X, \underline{M^\bullet/I^kM^\bullet})$$

Namely, choose any isomorphism

$$\underline{M^\bullet/I^nM^\bullet} \to K \otimes^{\mathrm{L}}_\Lambda \Lambda/I^n$$

induces similar isomorphisms module I^m and I^k and we see that the assertion is true. Thus to prove the lemma it suffices to show that the system $H^0(X, \underline{M^\bullet/I^nM^\bullet})$ has Mittag-Leffler. Since taking sections over X is exact, it suffices to prove that the system of Λ-modules

$$H^0(M^\bullet/I^nM^\bullet)$$

has Mittag-Leffler. Set $A = \Lambda^\wedge$ and consider the spectral sequence

$$\mathrm{Tor}^A_{-p}(H^q(M^\bullet), A/I^nA) \Rightarrow H^{p+q}(M^\bullet/I^nM^\bullet)$$

By More on Algebra, Lemma [original label: `more-algebra-lemma-tor-strictly-pro-zero`] the pro-systems $\{\mathrm{Tor}^A_{-p}(H^q(M^\bullet), A/I^nA)\}$ are zero for $p > 0$. Thus the pro-system $\{H^0(M^\bullet/I^nM^\bullet)\}$ is equal to the pro-system $\{H^0(M^\bullet)/I^nH^0(M^\bullet)\}$ and the lemma is proved. □

Lemma 11.6. Let X be a connected scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. If K is in $D_{cons}(X, \Lambda)$ such that $K \otimes_\Lambda \underline{\Lambda/I}$ has locally constant cohomology sheaves, then K is adic lisse (Definition 11.4).

Proof. Write $K_n = K \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n}$. We will use the results of Lemma 11.2 without further mention. By Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-locally-constant-bounded`] we see that K_n has locally constant cohomology sheaves for all n. We have $K_n = \epsilon^{-1}L_n$ some L_n in $D_{ctf}(X_{étale}, \Lambda/I^n)$ with locally constant cohomology sheaves. By Étale Cohomology, Lemma [original label: `etale-cohomology-lemma-connected-ctf-locally-constant`] there exist perfect $M_n \in D(\Lambda/I^n)$ such that L_n is étale locally isomorphic to $\underline{M_n}$. The maps $L_{n+1} \to L_n$ corresponding to $K_{n+1} \to K_n$ induces isomorphisms $L_{n+1} \otimes^{\mathrm{L}}_{\Lambda/I^{n+1}} \underline{\Lambda/I^n} \to L_n$. Looking locally on X we conclude that there exist maps $M_{n+1} \to M_n$ in $D(\Lambda/I^{n+1})$ inducing isomorphisms $M_{n+1} \otimes_{\Lambda/I^{n+1}} \Lambda/I^n \to M_n$, see Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-locally-constant-map`]. Fix a choice of such maps. By More on Algebra, Lemma [original label: `more-algebra-lemma-Rlim-perfect-gives-complete`] we can find a finite complex $M^\bullet$ of finite projective $\Lambda^\wedge$-modules and isomorphisms $M^\bullet/I^nM^\bullet \to M_n$ in $D(\Lambda/I^n)$ compatible with the transition maps. To finish the proof we will show that K is locally isomorphic to

$$\underline{M^\bullet}^\wedge = \lim \underline{M^\bullet/I^nM^\bullet} = R\lim \underline{M^\bullet/I^nM^\bullet}$$

Let $E^\bullet$ be the dual complex to $M^\bullet$, see More on Algebra, Lemma [original label: `more-algebra-lemma-dual-perfect-complex`] and its proof. Consider the objects

$$H_n = R\mathcal{H}\mathrm{om}_{\Lambda/I^n}(\underline{M^\bullet/I^nM^\bullet}, K_n) = \underline{E^\bullet/I^nE^\bullet} \otimes^{\mathrm{L}}_{\Lambda/I^n} K_n$$

of $D(X_{pro\text{-}étale}, \Lambda/I^n)$. Modding out by I^n defines a transition map $H_{n+1} \to H_n$. Set $H = R\lim H_n$. Then H is an object of $D_{cons}(X, \Lambda)$ (details omitted) with $H \otimes^{\mathrm{L}}_\Lambda \underline{\Lambda/I^n} = H_n$. Choose a covering $\{W_t \to X\}_{t \in T}$ with each W_t affine and weakly contractible. By our choice of $M^\bullet$ we see that

$$\begin{aligned} H_n|_{W_t} &\cong R\mathcal{H}\mathrm{om}_{\Lambda/I^n}(\underline{M^\bullet/I^nM^\bullet}, \underline{M^\bullet/I^nM^\bullet}) \\ &= \underline{\mathrm{Tot}(E^\bullet/I^nE^\bullet \otimes_{\Lambda/I^n} M^\bullet/I^nM^\bullet)} \end{aligned}$$

Thus we may apply Lemma 11.5 to $H = R\lim H_n$. We conclude the system $H^0(W_t, H_n)$ satisfies Mittag-Leffler. Since for all $n \gg 1$ there is an element of $H^0(W_t, H_n)$ which maps to an isomorphism in

$$H^0(W_t, H_1) = \mathrm{Hom}(\underline{M^\bullet/IM^\bullet}, K_1)$$

we find an element $(\varphi_{t,n})$ in the inverse limit which produces an isomorphism mod I. Then

$$R\lim \varphi_{t,n} : \underline{M^\bullet}^\wedge|_{W_t} = R\lim \underline{M^\bullet/I^nM^\bullet}|_{W_t} \longrightarrow R\lim K_n|_{W_t} = K|_{W_t}$$

is an isomorphism. This finishes the proof. □

Proposition 11.7. Let X be a Noetherian scheme. Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal. Let K be an object of $D_{cons}(X, \Lambda)$. Then K is adic constructible (Definition 11.4).

Proof. This is a consequence of Lemma 11.6 and the fact that a Noetherian scheme is locally connected (Topology, Lemma [original label: `topology-lemma-locally-Noetherian-locally-connected`]), and the definitions. □

12 Proper base change

In this section we explain how to prove the proper base change theorem for derived complete objects on the pro-étale site using the proper base change theorem for étale cohomology following the general theme that we use the pro-étale topology only to deal with "limit issues" and we use results proved for the étale topology to handle everything else.

Theorem 12.1. Let $f : X \to Y$ be a proper morphism of schemes. Let $g : Y' \to Y$ be a morphism of schemes giving rise to the base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

Let Λ be a Noetherian ring and let $I \subset \Lambda$ be an ideal such that Λ/I is torsion. Let K be an object of $D(X_{pro\text{-}étale})$ such that

1. K is derived complete, and
2. $K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$ is bounded below with cohomology sheaves coming from $X_{étale}$,
3. Λ/I^n is a perfect Λ-module[7].

Then the base change map

$$Lg^*_{comp} Rf_* K \longrightarrow Rf'_* L(g')^*_{comp} K$$

is an isomorphism.

Proof. We omit the construction of the base change map (this uses only formal properties of derived pushforward and completed derived pullback, compare with Cohomology on Sites, Remark [original label: `sites-cohomology-remark-base-change`]). Write $K_n = K \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n}$. By Lemma 2.1 we have $K = R\lim K_n$ because K is derived complete. By Lemmas 2.2 and 2.1 we can unwind the left hand side

$$Lg^*_{comp} Rf_* K = R\lim Lg^*(Rf_* K) \otimes^{\mathrm{L}}_{\Lambda} \underline{\Lambda/I^n} = R\lim Lg^* Rf_* K_n$$

the last equality because Λ/I^n is a perfect module and the projection formula (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-projection-formula`]). Using Lemma 2.2 we can unwind the right hand side

$$Rf'_* L(g')^*_{comp} K = Rf'_* R\lim L(g')^* K_n = R\lim Rf'_* L(g')^* K_n$$

the last equality because Rf'_* commutes with $R\lim$ (Cohomology on Sites, Lemma [original label: `sites-cohomology-lemma-Rf-commutes-with-Rlim`]). Thus it suffices to show the maps

$$Lg^* Rf_* K_n \longrightarrow Rf'_* L(g')^* K_n$$

are isomorphisms. By Lemma 1.8 and our second condition we can write $K_n = \epsilon^{-1} L_n$ for some $L_n \in D^+(X_{étale}, \Lambda/I^n)$. By Lemma 5.1 and the fact that ϵ^{-1} commutes with pullbacks we obtain

$$Lg^* Rf_* K_n = Lg^* Rf_* \epsilon^* L_n = Lg^* \epsilon^{-1} Rf_* L_n = \epsilon^{-1} Lg^* Rf_* L_n$$

and

$$Rf'_* L(g')^* K_n = Rf'_* L(g')^* \epsilon^{-1} L_n = Rf'_* \epsilon^{-1} L(g')^* L_n = \epsilon^{-1} Rf'_* L(g')^* L_n$$

(this also uses that L_n is bounded below). Finally, by the proper base change theorem for étale cohomology (Étale Cohomology, Theorem [original label: `etale-cohomology-theorem-proper-base-change`]) we have

$$Lg^* Rf_* L_n = Rf'_* L(g')^* L_n$$

(again using that L_n is bounded below) and the theorem is proved. □

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[7] This assumption can be removed if K is a constructible complex, see [1].